Note

In the above example, the values of x and y which satisfy one equation should also satisfy all the other equations. In other words, the equations are to be satisfied by the same values of x and y simultaneously. If such values of x and y exist, then they are said to form a **solution** for the system of linear equations. In the three equations, x and y appear in first degree only. Hence they are said to form a system of linear equations in two unknowns x and y. They are also called **simultaneous linear equations** in two unknowns x and y. The system has three linear equations in two unknowns x and y.

The equations represent three straight lines in two-dimensional analytical geometry.
In this section, we develop methods using matrices to find solutions of systems of linear equations.

1.4.2 System of Linear Equations in Matrix Form

A system of m linear equations in n unknowns is of the following form:

$$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1, \\ &a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2, \\ &\cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ &a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_m, \end{aligned} \qquad \ldots (1)$$

where the coefficients $a_{ij}, i = 1, 2, \cdots, m;\ j = 1, 2, \cdots, n$ and $b_k, k = 1, 2, \cdots, m$ are constants. If all the b_k's are zeros, then the above system is called a **homogeneous system** of linear equations. On the other hand, if at least one of the b_k's is non-zero, then the above system is called a **non-homogeneous system** of linear equations. If there exist values $\alpha_1, \alpha_2, \cdots, \alpha_n$ for $x_1, x_2, \cdots, x_n$ respectively which satisfy every equation of (1), then the ordered $n-$tuple $(\alpha_1, \alpha_2, \cdots, \alpha_n)$ is called a **solution** of (1).

The above system (1) can be put in a matrix form as follows:

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$ be the $m \times n$ matrix formed by the coefficients of $x_1, x_2, x_3, \cdots, x_n$. The first row of A is formed by the coefficients of $x_1, x_2, x_3, \cdots, x_n$ in the same order in which they occur in the first equation. Likewise, the other rows of A are formed. The first column is formed by the coefficients of x_1 in the m equations in the same order. The other columns are formed in a similar way.

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be the $n \times 1$ order column matrix formed by the unknowns $x_1, x_2, x_3, \cdots, x_n$.

Let $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ be the $m \times 1$ order column matrix formed by the right-hand side constants $b_1, b_2, b_3, \cdots, b_m$.

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